



Thoshith R R 250701797-2025-CSE ▾

T2

Started on	Friday, 14 August 2026, 3:46 PM
State	Finished
Completed on	Friday, 14 August 2026, 4:00 PM
Time taken	13 mins 55 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  void function(int n) {
4      int i = 1;
5      int s = 1;
6      int count = 0;
7      count++;
8      count++;
9
10     while (s <= n) {
11         count++;
12         i++;
13         count++;
14         s += i;
15         count++;
16     }
17     count++;
18
19     printf("%d", count);
20 }
21
22 int main() {
23     int n;
24     scanf("%d", &n);
25     function(n);
26     return 0;
27 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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